
EXERCISES

AI 504 exam practice

AI504 — Knowledge Representation

Simon Holm
May, 2026



DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

Multiple choice questions	3
Short-answer questions	5
Build your own logic	8

Multiple choice questions

1

- (a) ✓
- (b) ✓
- (c) ✓
- (d) ✓
- (e) ✕

2

- (a) ✓
- (b) ✕
- (c) ✕
- (d) ✕
- (e) ✕
- (f) ✕

3

- (a) ✓
- (b) ✓
- (c) ✓
- (d) ✓

4

- (a) ✓
- (b) ✕

5

- (a) ✕
- (b) ✕
- (c) ✓
- (d) ✓

6

- (a) ✕
- (b) ✓
- (c) ✕
- (d) ✕

7

- (a) ✕

- (b) ✕
- (c) ✕
- (d) ✓

8

- (a) ✓
- (b) ✓
- (c) ✓
- (d) ✕

9

- (a) ✓
- (b) ✓
- (c) ✕
- (d) ✕

10

- (a) ✕
- (b) ✕
- (c) ✕
- (d) ✓

11

- (a) ✕
- (b) ✓
- (c) ✕
- (d) ✕

12

- (a) ✕
- (b) ✕
- (c) ✕
- (d) ✓

Short-answer questions

1

Let ϕ be a contradiction such that $\phi = p \wedge \neg p$

Then

$$\mathcal{M} \models \neg(p \wedge \neg p) = \neg p \vee p \quad \top$$

$$\mathcal{M}^* \models p \wedge \neg p \quad \perp$$

$$\mathcal{M} \models \neg\phi \iff \mathcal{M}^* \models \phi \quad \perp$$

2

Let Γ, ϕ be arbitrary. Prove $\Gamma \vdash \phi$ then $\Gamma^\dagger \vdash \phi^\dagger$

Let $\Gamma \vdash \phi$ This means that there exists a proof tree that proves ϕ by Γ .

$$\exists \mathcal{D} \in \text{tree}_{\Gamma, \phi} : \Gamma^\dagger \vdash \phi^\dagger$$

$$(\forall \mathcal{D} \in \text{tree}_{\Gamma, \phi}). \Gamma \vdash \phi \Rightarrow \Gamma^\dagger \vdash \phi^\dagger$$

3

No

given that $\mathcal{M}_1 \models (p, q)$ and $\mathcal{M}_2 \models (q, \neg p)$ Then

$$\mathcal{M}_3 \not\models (p, \neg p)$$

4

$$p \Rightarrow q$$

5

$$P = \{p, q\}$$

$$\Gamma = \{\text{All } p \rightarrow q, \text{All } q \rightarrow p \text{ (see all } p)\}$$

$$\llbracket p \rrbracket = \{p\}$$

$$\llbracket q \rrbracket = \{q, p\}$$

$$\llbracket \text{see} \rrbracket = \{(q, p), (p,)\}$$

6

if $t = p$, then

$$f(t) = t(x)$$

if $t = v$ all t , then

$$F(t) = (\forall y)(f(T) \rightarrow r(x, y))$$

Proof by induction

Goal: For any term t : $\llbracket t \rrbracket = \llbracket f(t) \rrbracket$.

1 **Base case**

2 | Let $t = p$

3 | Since $\llbracket p(x) \rrbracket$ is the set of all element that satisfy $p(x)$,
then

4 |
$$\llbracket p \rrbracket = \llbracket f(p) \rrbracket = \llbracket p(x) \rrbracket$$

5 **Inductive hypothesis**

6 | Assume that for any subterm t' $\llbracket t' \rrbracket = \llbracket f(t') \rrbracket$

7 **Inductive step**

8 | Let t be an arbitrary term on the form $t = r$ all t' , then

9 |
$$\llbracket t \rrbracket = \{x \in M : \forall y \in \llbracket t' \rrbracket. (x, y) \in \llbracket r \rrbracket\}$$

10 |
$$\llbracket f(t) \rrbracket = \{x \in M : \forall y \in \llbracket f(t') \rrbracket. (x, y) \in \llbracket r \rrbracket\}$$

11 | By ***IH***

12 |
$$\llbracket t \rrbracket = \llbracket f(t) \rrbracket$$

7

$$\llbracket z \rrbracket = \emptyset$$

$$\llbracket \ell \text{ all } z \rrbracket = \mathbb{Z}$$

$$\llbracket \ell \text{ all } \ell \text{ all } z \rrbracket = \{x \in \mathbb{Z} : \forall y \in \llbracket \ell \text{ all } z \rrbracket. (x, y) \in \llbracket r \rrbracket\} = \emptyset$$

It flips

8

Definition of tableau rules for $\Rightarrow$ and $\Leftrightarrow$

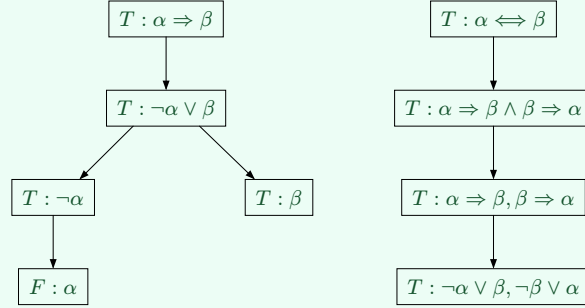
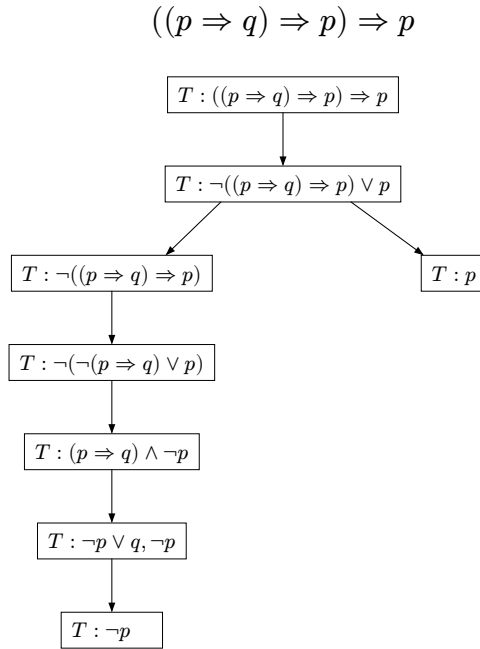


Figure 1: Rules for tableau expansion (propositional logic)

9



This mean that for $((p \Rightarrow q) \Rightarrow p) \Rightarrow p$ to be true then $p \vee \neg p$ this is a tautology

10

If $(\forall x, y)(L(x, y) \Rightarrow L(y, x))$, then $(\forall x, y)(L(y, x) \Rightarrow L(x, y))$

That means

$$(\forall x, y)(L(x, y) \Leftrightarrow L(y, x))$$

11

$$\mathcal{M} = (\emptyset, \llbracket \cdot \rrbracket)$$

12

$$\exists n \in \mathbb{N}, \forall x \in \{y \in \mathbb{N} \mid y > n\}. \text{test}(x)$$

13

$$x, y, z \in \mathbb{R}. \exists x \in \mathbb{N}. x < y < z$$

$$\forall x, y, z \in \mathbb{Z}. \exists n, m, k \in \mathbb{Z}. (x + 360n) - (z + 360k) < 360 \wedge$$

$$(z + 360k) < (y + 360m) \wedge$$

$$(y + 360m) < (x + 360n) \wedge$$

$$(\forall x, y \in \mathbb{Z})(\exists z \in \mathbb{Z}) : (x \bmod 360) \leq (z \bmod 360) \leq (y \bmod 360)$$

$$(\forall x, y \in \mathbb{Z})(\exists z, n, m, k \in \mathbb{Z}) : (x + n360) \leq (z + m360) \leq (y + k360)$$

Build your own logic

1

$$(a) \mathcal{M} = \{\Gamma, \llbracket \cdot \rrbracket\}$$

$$\forall p \in P : \llbracket p \rrbracket_{\mathcal{M}} \subseteq M$$

$$\forall n \in N : \llbracket n \rrbracket_{\mathcal{M}} \subseteq M$$

Let

$$\mathcal{M} \models \phi$$

$$\phi = \begin{cases} \llbracket p \rrbracket_{\mathcal{M}} \subseteq \llbracket q \rrbracket_{\mathcal{M}} & \text{if } \phi = \text{All } p \text{ } q \\ \llbracket n \rrbracket_{\mathcal{M}} \in \llbracket p \rrbracket_{\mathcal{M}} & \text{if } \phi = n \text{ is a } p \end{cases}$$

(b)

$$\frac{n \text{ is a } p \quad \text{All } p \text{ } q \text{ ISA}}{n \text{ is a } q}$$

- Soundness

Given the definition of a sentence ϕ

$$\llbracket n \rrbracket_{\mathcal{M}} \in \llbracket p \rrbracket_{\mathcal{M}} \subseteq \llbracket q \rrbracket_{\mathcal{M}}$$

$$\llbracket n \rrbracket_{\mathcal{M}} \in \llbracket q \rrbracket_{\mathcal{M}}$$

2

3

4

(a)

1. *fewer things*, since $\models^s$ is smaller
- 2.

$$\text{All } p \text{ } r \text{ all } q \Leftrightarrow \text{All } q \text{ } r \text{ all } p$$